

Bayesian quantile correction and synthesis for climate products

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QUESTION

Correction before synthesis

How can historical archives estimate source corrections before forecasts with different horizons are blended into one predictive distribution?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge, modeled on the $\log(1 + x)$ scale with x in m^3/s . Before each forecast origin, USGS observations and retrospective ECMWF/-GloFAS and NOAA/NWS products identify time-varying source corrections. After that origin, issued ensembles and exogenous information produce source-adjusted predictive flow quantiles.



USGS observations
15-min public gauge data; daily log-flow target for state updating and held-out verification.



ECMWF GloFAS
retrospective alignment; daily 51-member forecast guidance to 28 days.

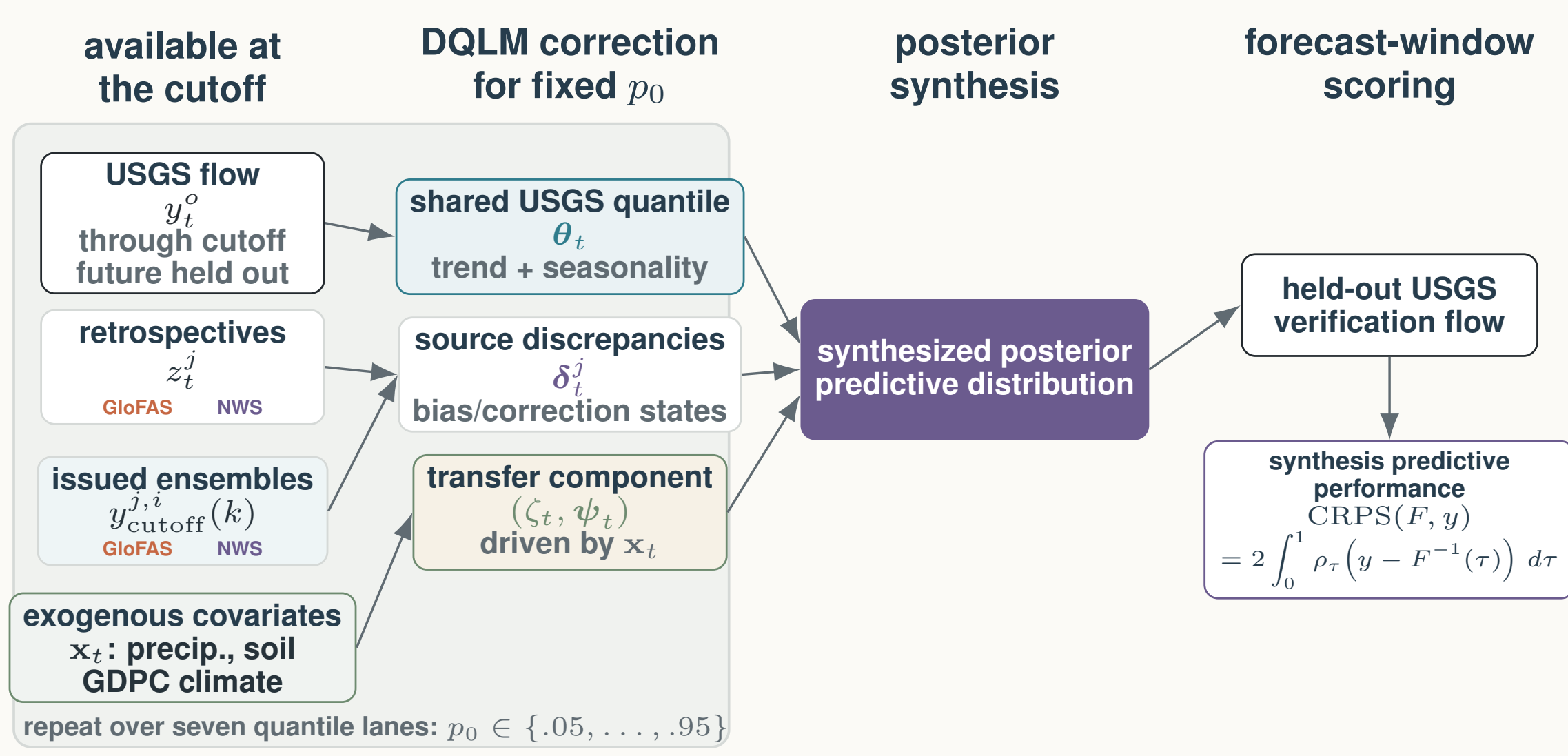


NOAA NWS
retrospective alignment; hourly forecasts evaluated on common daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

MODEL

Main Workflow



Source-aware DQLM workflow. For each target quantile p_0 , pre-origin USGS observations and retrospective product histories identify a latent river-flow quantile and source-specific correction states. After the cutoff, issued ensemble forecasts and staged exogenous covariates propagate corrected forecast-window quantiles. The seven fitted quantile lanes are then synthesized into one posterior predictive distribution and scored only with held-out USGS observations.

Dynamic Quantile Linear Model

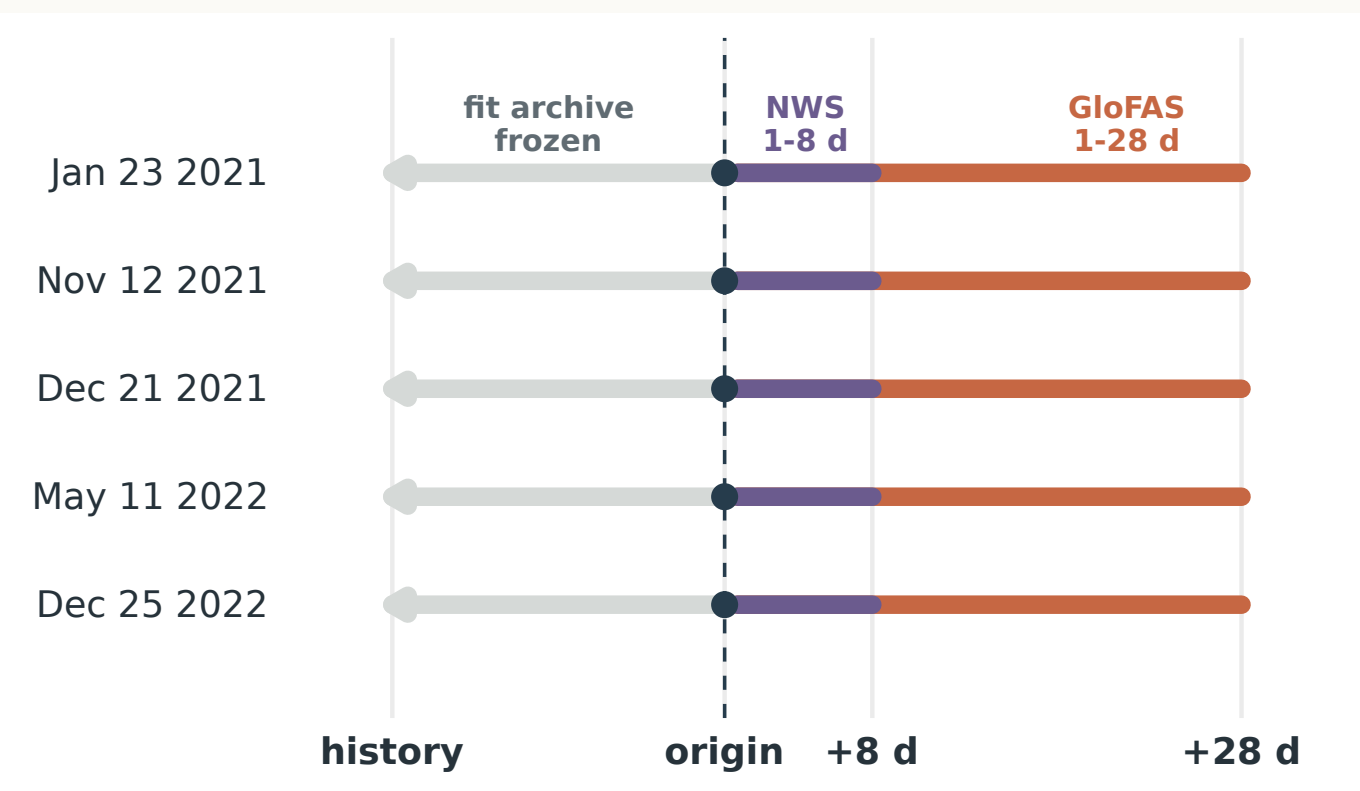
For each target quantile p_0 , let T denote the forecast origin/cutoff. The Dynamic Quantile Linear Model (DQLM) links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Product-family discrepancies learn source corrections before the origin; the same source channels, with forecast-window covariates x_{T+k} , then define corrected predictive quantiles after the origin.

$$\begin{aligned} \text{Observed measurements: } y_t^o &\sim \text{exAL}_{p_0}(\mathbf{F}_t^o \theta_t + \zeta_t, \sigma^o, \gamma^o), & 1 \leq t \leq T, \\ \text{Retrospective products: } z_t^j &\sim \text{exAL}_{p_0}(\mathbf{F}_t^j(\theta_t + \delta_t^j) + \zeta_t, \sigma^j, \gamma^j), & j \in \mathcal{J}, 1 \leq t \leq T, \\ \text{Forecast products: } y_{T+k}^{j,k} &\sim \text{exAL}_{p_0}(\mathbf{F}_{T+k}^j(\theta_{T+k} + \delta_{T+k}^j + \zeta_{T+k}, \sigma^j, \gamma^j), & j \in \mathcal{J}, i \in \mathcal{I}_j, k \in \mathcal{K}_j, \\ \text{Trend and seasonal states: } \theta_t | \theta_{t-1} &\sim \mathcal{N}(\mathbf{G}_t \theta_{t-1}, \mathbf{W}_t^o), & 1 \leq t \leq T + K_{\max}, \\ \text{Source correction states: } \delta_t^j | \delta_{t-1}^j &\sim \mathcal{N}(\mathbf{G}_t^j \delta_{t-1}^j, \mathbf{W}_t^j), & j \in \mathcal{J}, 1 \leq t \leq T + K_{\max}, \\ \text{Exogenous adjustment: } \zeta_t | \zeta_{t-1}, \psi_{t-1} &\sim \mathcal{N}(\lambda \zeta_{t-1} + \mathbf{x}_t^* \psi_{t-1}, \mathbf{W}_t^c), & 1 \leq t \leq T + K_{\max}, \\ \text{Covariate-adjustment states: } \psi_t | \psi_{t-1} &\sim \mathcal{N}(\psi_{t-1}, \mathbf{W}_t^c), & 1 \leq t \leq T + K_{\max}. \end{aligned}$$

exAL_{p_0} is the extended asymmetric Laplace working likelihood for target quantile p_0 ; σ and γ are source-specific scale and skewness terms. T is the forecast origin, j a product family, i an ensemble member, k a lead, and $K_{\max}$ the longest horizon. The shared quantile state is θ_t ; source discrepancies/corrections are δ_t^j ; and (ζ_t, ψ_t) is the exogenous-adjustment block driven by covariates x_t , such as precipitation, soil moisture, and the GDPC climate-index summary. Component discounting controls the evolution variances.

VALIDATION

Rolling-origin holdout design



Strict temporal holdout. At each origin, fitting uses only USGS flow and retrospective products available through that date. Issued forecasts and forecast-window exogenous inputs enter after the origin; future USGS observations are reserved for scoring. Main CRPS uses GloFAS-supported days 1–28, while NWS is compared over common leads 1–8.

INFERENCE

Scalable Bayesian quantile fitting

Seven independent Dynamic Quantile Linear Model (DQLM) lanes target seven levels: 0.05, 0.20, 0.35, 0.50, 0.65, 0.80, and 0.95. The fitted quantile forecasts are checked for crossing in the application and combined into the posterior predictive distribution scored by CRPS.

7 quantile lanes **≤ 12,995** USGS archive days **~ 2–4 h** fit + synthesis

Parallel quantile lanes

Each quantile level is fit as its own state-space model; the lanes can run concurrently before forecast quantiles are synthesized.

State distribution updates

Mean-field variational Bayes (MFVB) uses Kalman filtering and backward smoothing equations to update Gaussian state distributions within each latent-state block.

Non-conjugate updates

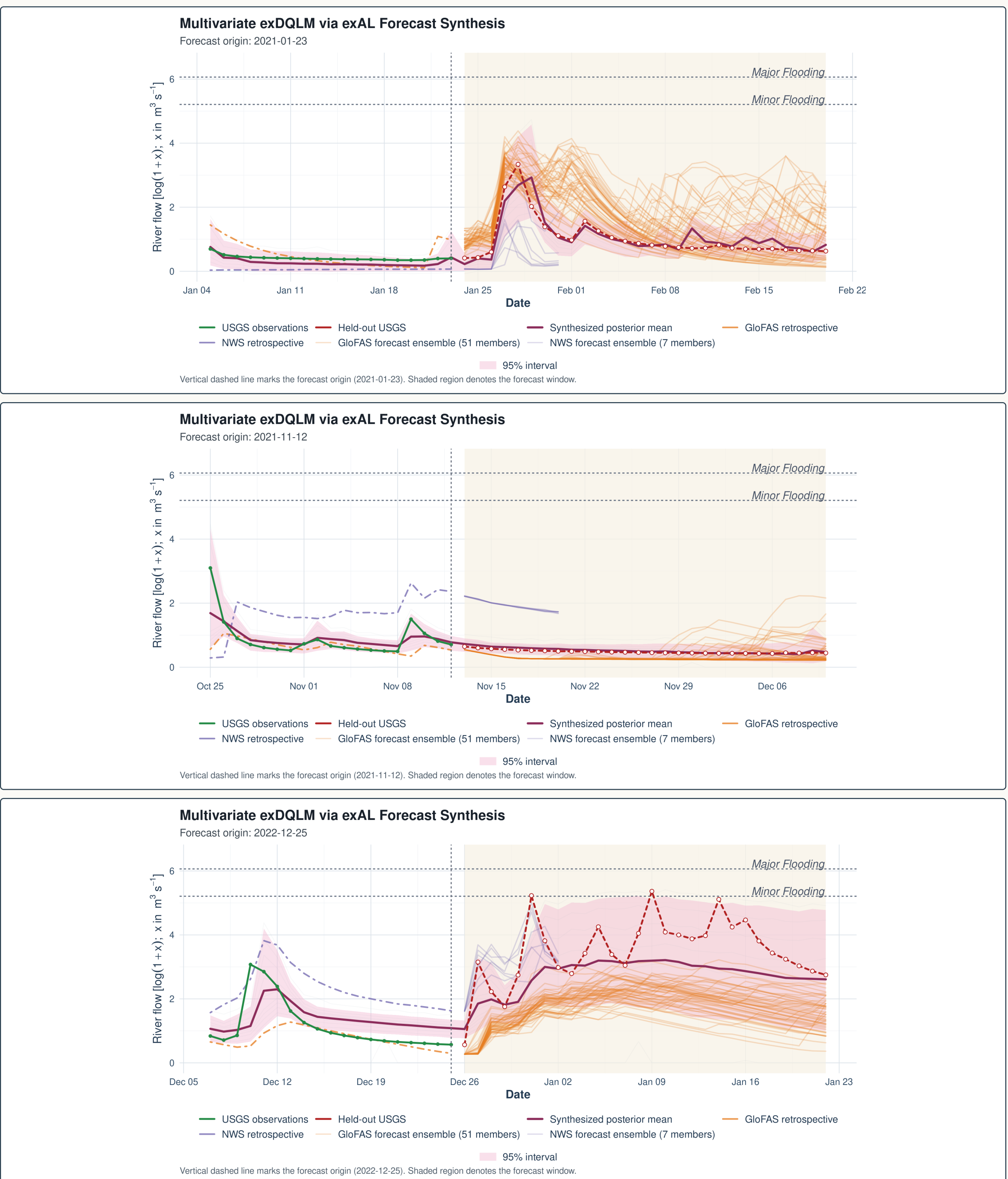
Variational Bayes with Laplace–Delta (VB-LD) uses local modes, curvature, and Delta-method moments for non-conjugate extended asymmetric Laplace (exAL) scale and skewness terms.

State evolution

Component discounting controls trend, multi-period seasonality, source corrections, and exogenous-covariate adjustments.

EXAMPLE

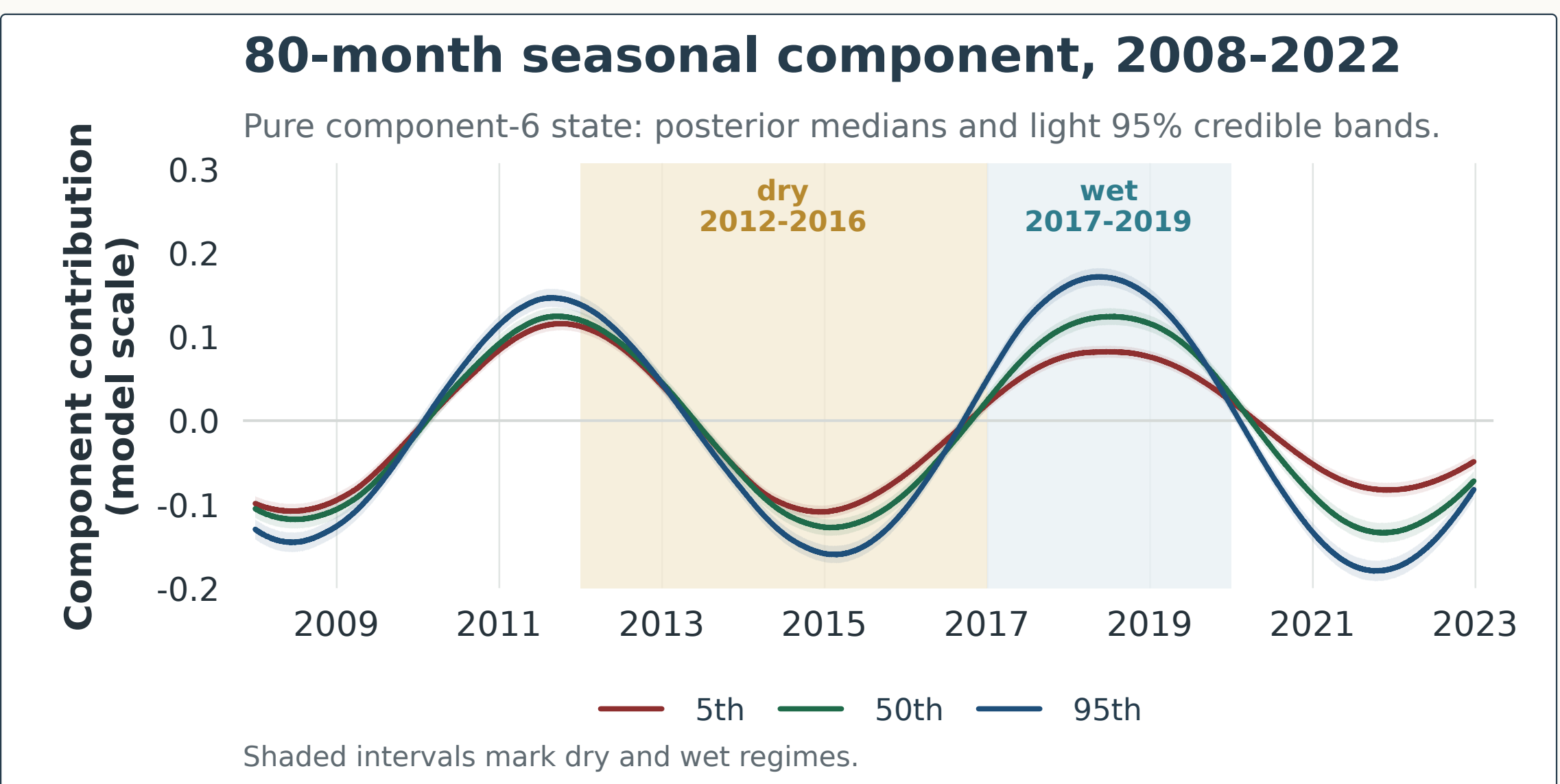
Three forecast-window examples



Origin-level illustrations. Three selected forecast windows show how corrected quantile forecasts combine into a posterior predictive envelope after the cutoff. The panels are illustrative checks of behavior at individual origins; the aggregate comparison remains the five-origin CRPS benchmark.

DYNAMICS

80-month seasonal component



Retrospective diagnostic. From the selected model at the 2022-12-25 origin. Lines show the isolated 80-month state component, without the trend component added, for the 5th, 50th, and 95th target-quantile lanes over 2008–2022; light ribbons show 95% credible bands and shaded intervals mark dry and wet regimes.

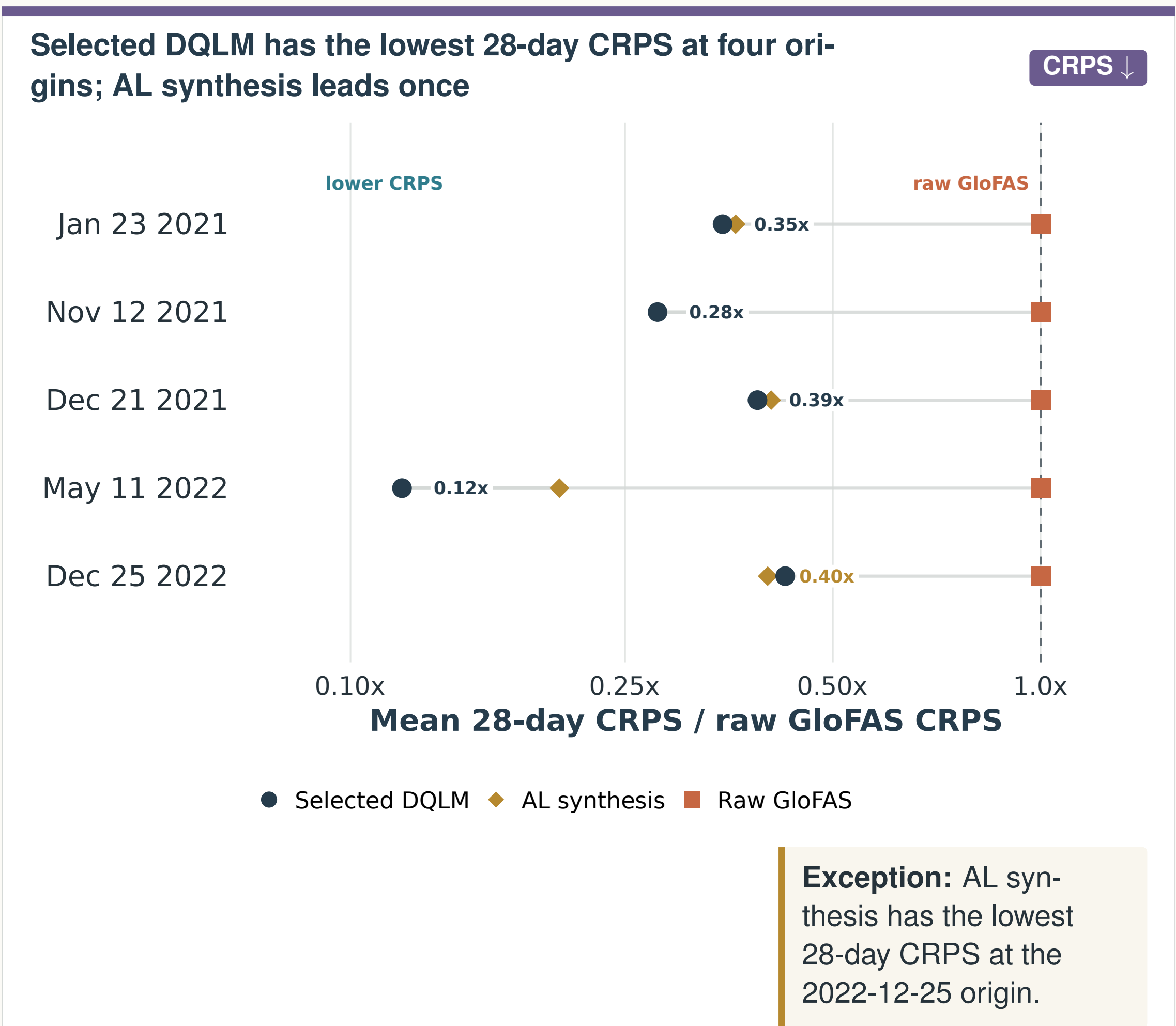
TAKEAWAYS

Main takeaways

- Historical archives learn source-specific corrections before forecast products are blended.
- Seven corrected quantile lanes are synthesized into one posterior predictive distribution.
- In this five-origin study, the selected DQLM has the lowest 28-day CRPS at four origins; 2022-12-25 favors AL synthesis.
- Fitted quantile dynamics remain interpretable: spread widens during wet high-flow periods.

RESULTS

28-day CRPS comparison

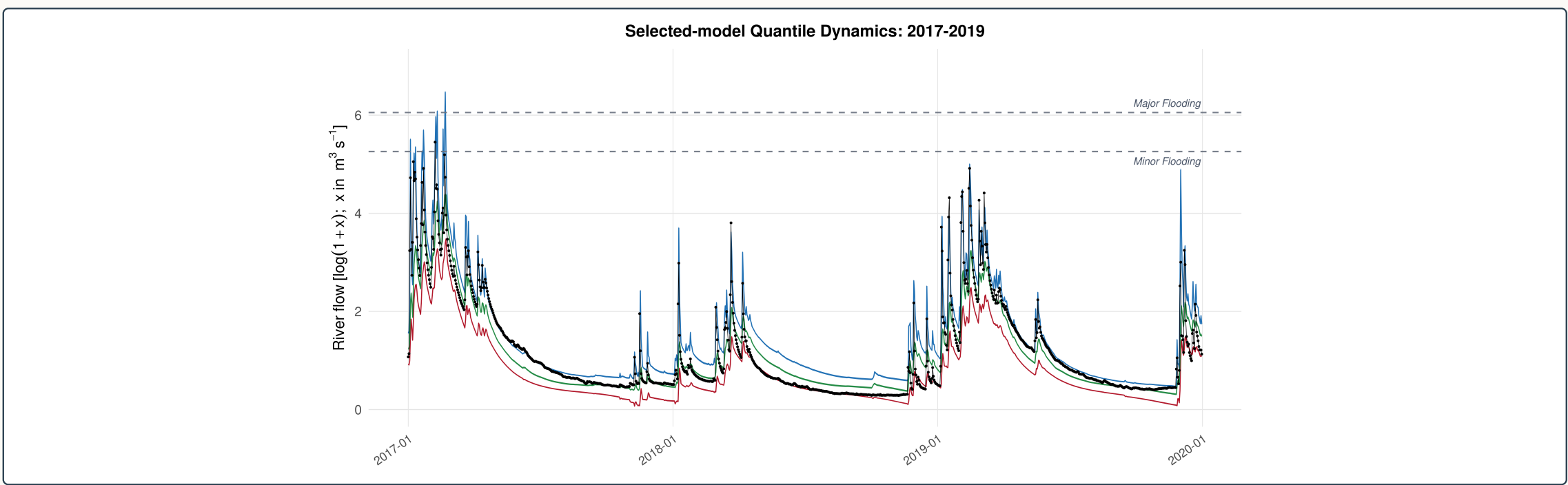


How to read this panel. Each point is mean forecast-window CRPS divided by raw GloFAS CRPS at the same origin. Values below 1.0 improve on the uncorrected 28-day GloFAS benchmark.

Values left of the raw GloFAS line indicate lower mean CRPS than raw GloFAS at that origin; raw NWS is evaluated over common leads 1–8 in the horizon-matched tables, not mixed into this 28-day GloFAS-supported benchmark.

FIT

Wet-period quantile fit



Historical fit diagnostic. Selected-model fitted quantile locations during the 2017–2019 wet period. The 5th, 50th, and 95th quantile fits are shown with observed USGS flow; wider upper-tail bands make winter high-flow behavior visible.

